

Citizen scientists can help advance the science and management of intermittent rivers and ephemeral streams

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Abstract

Intermittent rivers and ephemeral streams are the world's dominant type of river ecosystem and are becoming more common because of global change. However, the inclusion of intermittent rivers and ephemeral streams in water policies and management plans remains largely limited because monitoring schemes and tools are designed for perennial rivers. In the present article, we discuss how smartphone applications used by citizen scientists can quantify the extent and occurrence of intermittent rivers and ephemeral streams. We also introduce a new app, DRYRivERS, specifically designed to monitor intermittent rivers and ephemeral streams. After a year of use, we counted more than 3600 observations from more than 1900 river reaches across 19 countries and four continents. Through three case studies, we then show that citizen science can improve our knowledge of the prevalence of intermittent rivers and ephemeral streams in the landscape, enhance hydrological modeling and calibration, and guide managers in setting water abstraction restrictions. Together, our approach demonstrates how citizen science can be incorporated into environmental monitoring to better inform river management and policy.

Keywords: citizen science, intermittent rivers and ephemeral streams, smartphone app, hydrological modeling, water management

Decision-makers are faced with the pressing issues of addressing rapid and drastic environmental change to protect ecosystem health and human well-being. Environmental change occurs at different spatial scales and typically unfolds over long periods of time, requiring comprehensive and transdisciplinary approaches (Burke et al. 2017). Integrative assessments that fulfil environmental policy and include new ways of producing scientific data are necessary to achieve the goals set forth in a growing number of international treaties (e.g., Aichi Biodiversity targets, CBD 2013; the United Nations Sustainable Development Goals, Weisenborn 2018).

Intermittent rivers and ephemeral streams are the world's dominant type of river ecosystem (Messager et al. 2021), including more than 50% of the global hydrological network. This type of streams is becoming even more prevalent because of drivers of global change that include climate change (drought) and agricultural intensification with concomitant water demands (Datry et al. 2023). Intermittent rivers and ephemeral streams support diverse habitats, contributing significantly to aquatic–terrestrial biodiversity and major biogeochemical cycles. These functions, in turn, provide many ecosystem services crucial for society and human well-being, such as drinking water supply and sediment erosion control (Datry et al. 2018, Stubbington et al. 2020). However, the inclusion of these streams in water policies and management plans remain largely limited because monitoring schemes,

approaches, and tools are designed for perennial rivers, rarely reflecting the prevalence of intermittent rivers and ephemeral streams or their hydrological characteristics and biological communities (Datry et al. 2018).

The lack of knowledge about the spatial distribution and hydrology of intermittent rivers and ephemeral streams constitutes a significant hindrance to effectively managing these habitats. Existing hydrographic maps often misclassify stream types (Fritz et al. 2013) or do not include information on flow intermittence, because historical topographic surveys relied on single site visits, widespread *in situ* sensor installation is infeasible, and free-to-access satellite remote sensing provides data infrequently and at a coarse resolution (Senf 2022). Intermittent rivers and ephemeral streams are therefore not as likely to have stream gauges installed as perennial streams are. Drying can be driven by both natural factors (Larned et al. 2010, Hammond et al. 2021) and anthropogenic activities such as water abstraction, diversion, and flow regulation by dams (Datry et al. 2023). Flow alterations induced by climate change are also compounding the direct impacts of anthropogenic activities on river flows, which intensifies the dependencies of human–flow relationships and have substantial repercussions for societies, water managers, and policymakers (Anderson et al. 2019). Changes in public perception and greater education are therefore needed so that the public can understand the ecosystem services provided by the stream even when it is

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dry. Citizen science initiatives appear an appropriate solution to help locate, map, and monitor intermittent rivers and ephemeral streams at a fine spatial scale (Kampf et al. 2018) while raising people's awareness of dry and wet phases in river networks.

Hydrology historically lagged behind other disciplines in adopting citizen science (Buytaert et al. 2014) until 2010, when CrowdHydrology (Lowry and Fienen 2013, www.crowdhydrology.com) was established. The aim of CrowdHydrology is for citizen scientists to collect spatially distributed hydrologic data by sending observations of preinstalled gauges by text messages. Multiple hydrological projects have since emerged to capitalize on smartphones and their applications (Teacher et al. 2013). For instance, in 2016, Crowdwater (Seibert et al. 2019; <https://crowdwater.ch/en/start>) revolutionized citizen science in ecohydrology with the development of virtual gauges, which enabled users to measure stream level with a digital gauge inserted onto a photo. Other local initiatives such as RiuNet in Spain (www.ub.edu/fem/index.php/en/inici-riunet-en) and En Quête d'Eau in France (<https://enquetedeau.eaufrance.fr/accueil>) followed. However, although these projects collect a large number of contributions across the world, observations on intermittent rivers and ephemeral streams remain limited (less than one-third of the observations made in Europe by Crowdwater over the period from 17 June 2021 to 17 June 2022; www.crowdwater.ch). Recently, national-scale citizen-science programs such as StreamTracker in the United States (www.streamtracker.org) and DRYRivERS in Europe (www.dryver.eu/citizen-science/how-does-it-work) have emerged, specifically for targeting intermittent rivers and ephemeral streams.

By reporting drying events in rivers and streams through a smartphone application (hereafter, an *app*), citizen scientists can play a major role in mapping and monitoring hydrological states in intermittent rivers and ephemeral streams. This participation and the data generated helps overcome a major barrier to the effective management of intermittent rivers and ephemeral streams. In this study, we use data crowdsourced through smartphone apps to demonstrate how citizen science can help to serve regulatory bodies to monitor hydrological states across jurisdictional waters, to improve mapping of intermittent rivers and ephemeral streams in the landscape, and to document the current and future flow regimes of intermittent rivers and ephemeral streams. Finally, we discuss the potential benefits and limitations of using citizen science as a tool to tackle the complex challenges associated with intermittent rivers and ephemeral streams.

DRYRivERS, a dedicated app to monitor intermittent rivers and ephemeral streams

DRYRivERS is both a web and smartphone app developed within a European Research and Innovation Action project (DRYvER, www.dryver.eu, Detry et al. 2021). The web app is a real-time map of all recorded spots, whereas the smartphone app is made for quick and easy offline data recording in the field. DRYRivERS was first released for Android and iOS in September 2021 and April 2022, respectively, followed by four updates throughout 2021 and 2022. The app is currently available in more than 20 languages and therefore enables the georeferencing of observations worldwide. Two versions were developed: one for single users and one for groups and organizations. An organization account has extra options that, for example, allow the organization's administrators to invite new members to join their group or delete erroneous observations.

When first starting the smartphone app, the user has to create an account and browse through the intro tutorial that explains the

basic functionalities and the app interface. Additional guidance on how to use the app is provided through downloadable tutorials and on the project homepage. To log an observation at a new spot, the user clicks on "Add new spot or record" from the homepage (figure 1). After locating themselves through their GPS or manually, users have to record the hydrological state of the river or stream: flow, disconnected pools, or dry (figure 1). Finally, the citizen scientists are encouraged to name the spot and take a photo. While logging new observations, citizen scientists earn points and collect badges when reaching a certain number of observations (figure 1). Special badges for special days such as the World Water Day are worth more points than regular badges. This award system is designed to keep the citizen scientists entertained and to keep them active over the long term. The badging system was a very efficient way to collect repeated observations. Indeed, when comparing the 6-month period before and after the introduction of the badges, the number of sites having more than one record per month went up by nearly 83%, with a number of repeated observations that doubled (from maximum 13 observations before to a maximum of 26 observations after for a given site).

From the web app (www.dryver.eu/app; figure 2), it is possible to add and edit a previous or new observation using a desktop computer; explore and visualize the spatial distribution of observations overlaid on a satellite view or a terrain map; cluster the observations, filter by date, access basic statistics (e.g., the proportion of dry state among all the observations); and download all or a subset of the observations (e.g., filtered by date or including only a user's own observations).

In January 2023, DRYRivERS counted 1277 users among which 260 (i.e., 20.3%) were active users; that is, they made at least one observation. In total, 4204 observations were made, with 32% of these observations occurring within the case study watersheds of DRYvER (i.e., 1344 observations). In total, only 19% of the users were involved in DRYvER (i.e., scientists and water managers), but they recorded 59% of the observations. One-third of the citizen scientists (i.e., people not involved in the DRYvER project) only made one observation. For instance, in Hungary, where the spatial coverage is the best, 88.8% of the users are citizen scientists (laypeople, including students or scientists from a field other than water science or biology, and managers from biomonitoring agencies or natural parks) who recorded 998 (i.e., 58.4%) observations, and 11.2% are researchers in water-related topic and water managers (regardless of whether they are related to the DRYvER project or not) who made 711 observations (i.e., 41.6%).

Most of the users are occasional: 62% the users made between one and five observations since their registration. The top 10% of contributors supplied 69.4% of the observations. 40% of the citizen scientists kept engaged over extended periods (more than 6 months), with 8% of them becoming active users (more than 20 observations and used the app for longer than 6 months). Overall, 67% of the users made local observations (i.e., within a radius of 5 kilometers), with a peak of observations during the months of September and October.

How can drying observations help managers and scientists?

Our study presents three case studies to show that citizen science can guide managers in setting water abstraction restrictions, improve our knowledge of the prevalence of intermittent rivers and ephemeral streams in the landscape and, enhance hydrological modeling and calibration.

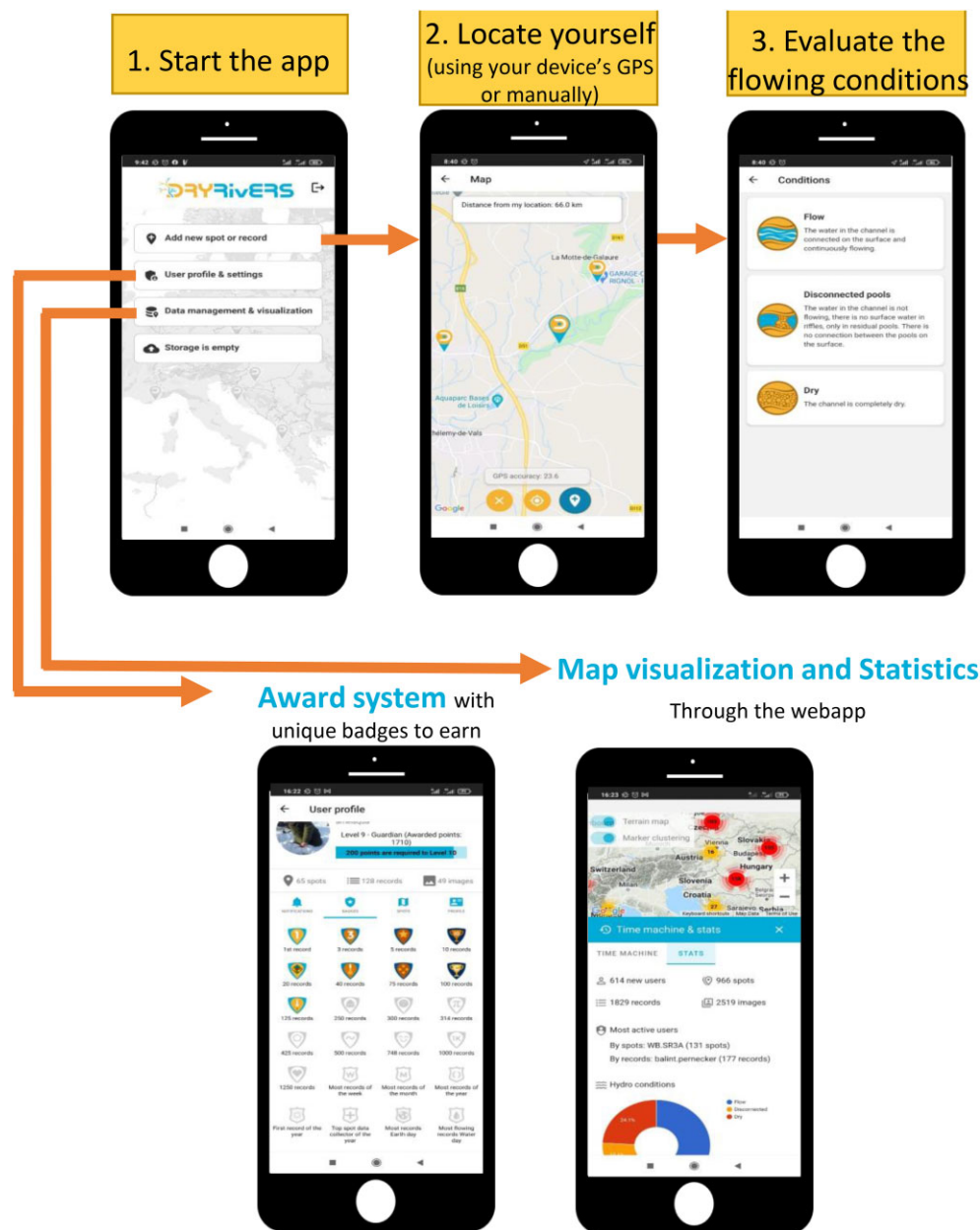


Figure 1. Screenshots of the DRYRivERS app displaying the two first steps needed to make an observation (numbered 2–3), along with the award system (badges and levels) and the tools developed for data visualization (direct link to the web app). (The online version of this image contains color.)

Case study I: Monitoring river hydrological state for regulatory purposes

The Syndicat de la Rivière d'Ain Aval et ses Affluents (SR3A) is a public authority responsible for managing the river Ain catchment, in France. As in many other regions, low-flow and dry periods are becoming increasingly frequent and severe in the Ain catchment (Sauquet et al. 2021). Consequently, SR3A has to report hydrological states to the French government, which, in turn, makes decisions about restrictions on water usage in the area. Because the area covered by SR3A is large (1700 square kilometers, including 1300 kilometers of river channels), managers seek opportunities to implement novel tools for monitoring hydrological states across river channels of the Ain catchment.

SR3A is using the DRYRivERS app rather than implementing water level *in situ* sensors. Sensors are expensive and time consuming to run; can be stolen, damaged, or washed out by

floods; and are considered less effective, because they do not help raise public awareness of intermittent rivers and ephemeral streams. Without citizen science smartphone apps, SR3A would not have been able to closely monitor the hydrological states of the catchment, because of both time and budget restrictions.

SR3A was actively involved in the development of DRYRivERS as beta testers, contributing significantly to tailoring the app to the needs of water managers. For example, SR3A has built a user network of a hundred representatives of local authorities (i.e., local officials, water managers) who are responsible to regularly monitor 128 checkpoints defined by SR3A itself (figure 3). Each representative has an account that is directly linked to the group account that SR3A supervises, allowing SR3A to track observations, delete erroneous data, etc.

SR3A also communicated widely about the smartphone app through watershed committees, public demonstrations on how

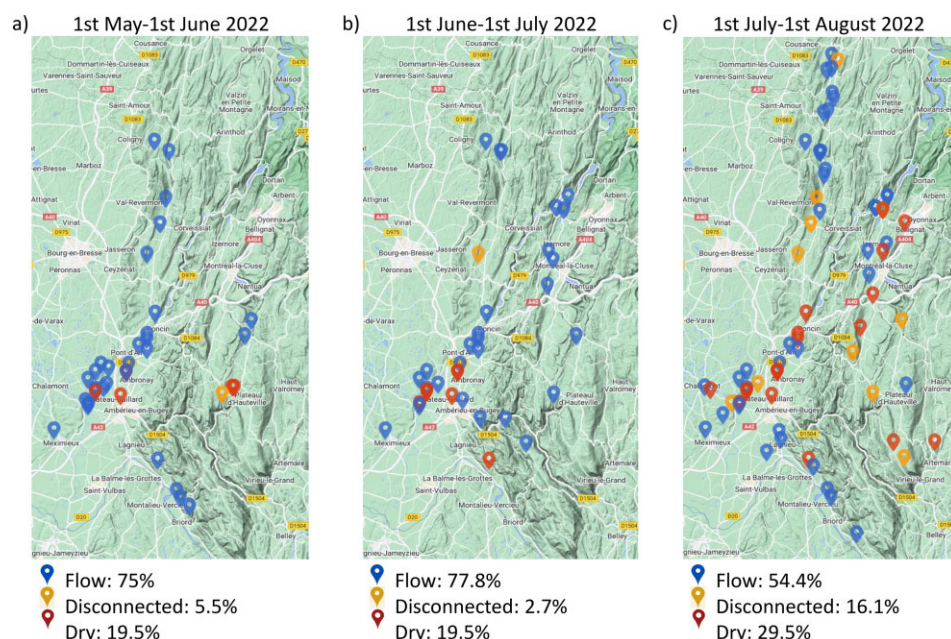


Figure 2. Evolution of river and stream drying in the area under SR3A's purview from 1 May 2022 (a) to 1 August 2022 (c). Flow states are color coded. (The online version of this image contains color.)

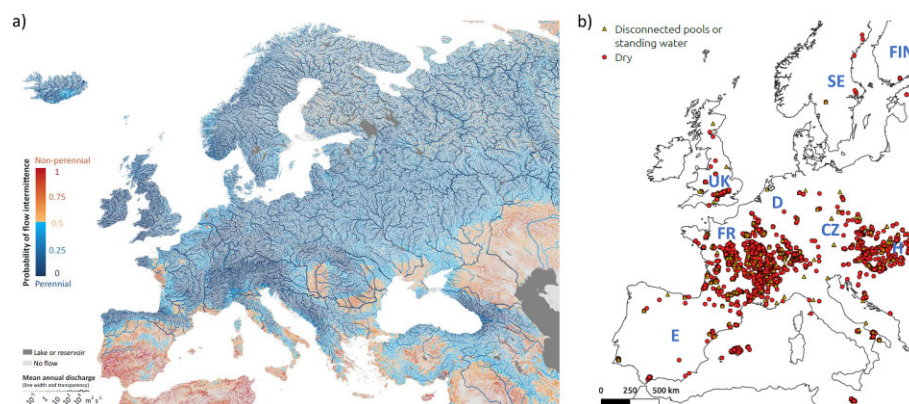


Figure 3. (a) Modeled distribution of IRES across Europe. Streams with a mean annual flow of at least 0.01 cubic meters per second were included in the data set of the study. Streams were considered as IRES (nonperennial, in red) when flow ceases for at least 1 day per year on average. Source: Courtesy of [Messager and colleagues \(2021\)](#). (b) Crowdsourced observations extracted from three different smartphone apps used in Europe (Crowdwater, DRYRIVERS and En Quête d'Eau). Dry states (the red dots) and disconnected pools and standing water states (the orange dots) are depicted. Abbreviations: CZ, Czech Republic; D, Germany; E, Spain; FIN, Finland; FR, France; H, Hungary; SE, Sweden; UK, United Kingdom. (The online version of this image contains color.)

to use the app, the creation of a dedicated webpage, and an online map of their checkpoints. SR3A is now hoping to enroll more users, such as members of fishing associations, who would collect observations outside of predefined checkpoints, an approach they judged as the most efficient to monitor intermittent rivers and ephemeral streams in their area. This initiative is motivated by the historical drought experienced in the region during summer of 2022 that raised general public awareness of river drying.

Case study II: Mapping intermittent rivers and ephemeral streams in the landscape

Drying events are not only observed in arid regions, as has been revealed both by modeled distribution of intermittent rivers and ephemeral streams (figure 3; [Messager et al. 2021](#)) and by crowd-sourced data. Intermittent rivers and ephemeral streams are indeed located in any climatic region and in any river network. Nev-

ertheless, crowdsourced observations were recorded in countries that were not identified as prone to drying by global hydrological models (figure 1; see [Messager et al. 2021](#) for a world map), such as the United Kingdom or central Europe (i.e., Germany, Czech Republic; figure 3a). Such observations could highlight the value of local knowledge (these sites were known to dry by local people and are picked up in local and regional hydrological models; i.e., chalk streams in England) that could complement or sometimes contradict hydrological models run at larger spatial scales.

Discrepancies between global hydrological models based on gauging stations ([Messager et al. 2021](#)) and crowdsourced data can also appear in regions where streams with a mean annual discharge over 0.1 cubic meters per second do not naturally cease to flow, meaning that the global models were unable to predict whether small streams (i.e., mean annual discharge below 0.1 cubic meters per second)—such as headwaters—dry up. This is a significant shortcoming, suggesting that the prevalence of

intermittent rivers and ephemeral streams is much higher when accounting for small streams. Indeed, crowdsourced observations in temperate regions showed that headwaters can also dry up for part of the year and outside of known drying river networks (e.g., in France or in Sweden; figure 3b). Previous studies have linked this drying to the small area in which headwaters usually drain (Costigan et al. 2016, Prancevic and Kirchner 2019) or to the lack of knowledge scientists may have when dense riparian cover prevents the use of remote sensing to locate new basins (Benstead and Leigh 2012, Sauquet et al. 2021).

Climate change and anthropogenic pressures (e.g., water storage and flow regulation by reservoirs for irrigation, flood control or hydropower) are also drivers of drying (Datry et al. 2023). Messenger and colleagues (2021) modeled the natural occurrence of drying, whereas crowdsourced data could reveal newly drying streams. Perennial streams can dry because of increasing frequency and severity of drought (Schindler et al. 1996, Spinoni et al. 2018), in response to, for example, dams or extreme climatic events such as the drought experienced in summer 2022 across Europe. Crowdsourced data therefore have the potential for supplementing data from gauging stations, favoring a more detailed map of intermittent rivers and ephemeral streams.

Case study III: Modeling hydrological states of intermittent rivers and ephemeral streams

Modeling drying events in river networks is a key step to understanding and conceptualizing intermittent rivers and ephemeral streams and to estimating hydrological patterns under future climate scenarios. However, hydrologists are facing a major problem: the vast majority of flow gauging stations are located on perennial rivers such that the observed input data to develop hydrological models and the resulting prediction of the prevalence of intermittent rivers and ephemeral streams are biased and cannot accurately replicate drying events (Snelder et al. 2013, Meerveld et al. 2020, Krabbenhoft et al. 2022).

In this case study, we assessed whether complementing standard flow gauging station data with crowdsourced observations of hydrological state to calibrate a hydrological model improved the model's predictions of flow intermittency. We modeled flow intermittency on the basis of 15,791 hydrological state observations across four catchments in four European countries (i.e., Finland, France, Hungary and Spain; supplemental figure S2a, supplemental table S1). Those catchments were chosen because they are part of the nine focal drying river networks studied by the DRYvER consortium (through which the DRYvER app was developed; Datry et al. 2021). DRYvER aims to model the effects of climate change on biodiversity, ecological integrity, and ecosystem services of intermittent rivers and ephemeral streams. Hydrological observations collected throughout the DRYvER project can be classified into three categories: gauging stations with long time series (including stations from the ONDE network designed by the French Biodiversity Agency to monitor intermittent rivers for the Albarine catchment only, <https://ofb.gouv.fr>), which represented the majority of available hydrological observations; data collected during field campaigns (i.e., photo traps, water level loggers); and crowdsourced data from smartphone apps (Crowdwater, DRYvERs). Gauging stations provide highly temporally resolved data (i.e., regular time steps—daily or monthly—over long periods) needed for the validation of hydrological models over a wide range of environmental and seasonal conditions. Field campaign data, such as photo traps, provide daily pictures of the streams and therefore collect a large amount of hydrological observations over a

relatively short period. Although they are both particularly useful to understanding the temporal dynamics of flow intermittence, they are also limited in space (between one and six stations per catchment; figure 4a, figure S2b–2e, supplemental table S2) and monitor only small fractions of the catchments.

In contrast, crowdsourced observations made through smartphone apps provide a very dense spatial coverage (21–76 monitored river sections per catchment; table S2) but with a lower temporal resolution (on average, three observations per monitored river section). The temporal resolution varied over time, from several records per day to records made every 6 months. Even for a given site, the temporal resolution can vary from 15 records a day to one record 4 days apart and then one record every 6 months (Velička Stream, Czech Republic, with an average of one record every 8 days, 65 records over 18 months in total). In total, crowdsourced data made up to 4% of all hydrological observations (figure 4b) but included more than 23% of the observed dry states (figure 4c) and more than 91% of the length of the observed river network in all four catchments (figure 4d) and provide unique information regarding the spatial patterns of intermittence in drying river networks.

The observed hydrological data were then used to train a flow intermittence model (the modeling method is fully described in supplement S1) in order to predict the hydrological states (dry or flow) at daily time steps for each river section of the four catchments. To assess the contribution of crowdsourced data to the modeling of flow intermittence, we compared the outputs of the model trained once with all types of observed data (i.e., crowdsourced, stations, field campaign) with the model trained without the crowdsourced data (i.e., stations, field campaign). The outputs of the model showed an increase of the performance of the model to reproduce dry events when the crowdsourced data were included in the training data set (figure 5a). This is particularly true in the Genal catchment, where almost all observations of dry states come from the crowdsourced data (figures 4b and 5a). In the Albarine catchment, crowdsourced data significantly helped reduce the misclassification rates (i.e., flowing events wrongly simulated as dry; figure 5a). The inclusion of crowdsourced data also had an impact on the simulated length of dry river network in the basins (figure 5b). In the Genal catchment, the model simulated almost no dry river sections when crowdsourced data were not taken into account, and in the Albarine catchment, the model simulated 17 kilometers less of dry rivers when the crowdsourced data were included.

In conclusion, crowdsourced observations improved the performance of the modeling of hydrological states of intermittent rivers and ephemeral streams, especially in catchments where hydrological stations are scarce or when field campaigns cannot be implemented.

Conclusions

Through the case studies presented in the present article, we demonstrated the importance of citizen science initiatives in advancing research and management of intermittent rivers and ephemeral streams. Crowdsourced data represent a new type of data in research on intermittent rivers and ephemeral streams, particularly for hydrological modeling. Most hydrological models are developed from daily gauging records and to date, crowdsourced observations have seldom been used for calibration or validation of such models (but see Jaeger et al. 2023). Crowdsourced observations of hydrological states only provide snapshots without knowing whether the observed state

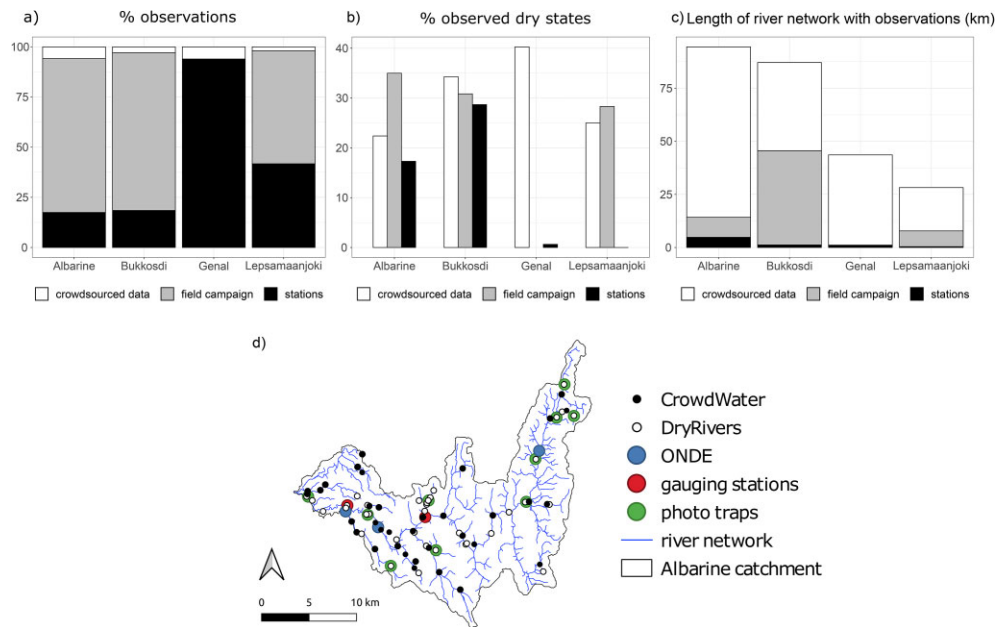


Figure 4. (a–c) Barplots showing key figures for observations from smartphone apps (in white) versus gauging stations (in black) and field campaigns (in grey) made within four different European catchments (Albarine, France; Bukkosdi, Hungary; Lepsamaanjoki, Finland; and Genal, Spain). The percentage of the total observations, the percentage of dry states among all the observations and length of observed river network were made from 1 January 2018 to 31 December 2021. (d) The location of observations coming from smartphone apps (Crowdwater and DRYRivers), field campaigns (photo traps) and hydrological stations (ONDE and gauging stations) within the Albarine catchment, France. Maps for other catchments can be found in supplemental figure S2. The general characteristics of the catchments are reported in supplemental table S1, and the exact figures supporting the barplots are in supplemental table S2. (The online version of this image contains color.)

represents the norm or exceptional conditions. Nevertheless, citizen science yields valuable observations where gauging data are currently lacking, allowing scientists to accurately predict the geographical distribution and the hydrological dynamics of intermittent rivers and ephemeral streams. Combining these two kinds of data is promising (Jaeger et al. 2023) and already suggests that intermittent rivers and ephemeral streams might be even more prevalent than currently estimated by global hydrological models (e.g., Messenger et al. 2021), paving the way to realistic predictions of flow dynamics under climate change scenarios.

Engaging citizens to observe flow dynamics of intermittent rivers and ephemeral streams is an excellent way of raising people's awareness on this type of rivers and streams. This approach also helps reconnect people to river and streams by emphasizing the importance of the water resources and any resulting restrictions that the authorities may implement in response to more frequent or more dramatic drying. However, knowing who the citizen scientists are is a difficult exercise, because data is scarce (e.g., no publicly available or published data for CrowdWater). One reason is that the EU General Data Protection Regulation does not allow personal information to be stored or used. In DRYRivers, the engagement of the general public was a real success, because, for example, 88.8% of the Hungarian users and 72.5% of the French users were citizens who were not related to the DRYVER project. These engagement rates from the general public are higher than the ones observed for En Quête d'Eau, for which 11.2% of the users fall into the general public class (Moitrier and Nowak 2022). Moreover, a small share of citizen scientists (i.e., 10%) contributed to a large share of observations (69.4%)—shares that are comparable to those previously observed in other citizen science projects (i.e., 10% of citizens making 71%–88% of observations, Sauermann and Franzoni 2015). These figures are very encouraging, because society considers that the only valuable

state of a river is when the stream flows (Leigh et al. 2019). Most people are not aware that intermittent rivers and ephemeral streams—often regarded as unaesthetic and even in a degraded state (Boulton 2014, Leigh et al. 2019, Cottet et al. 2022)—can also deliver valuable ecosystem services (Koundouri et al. 2017, Datry et al. 2023). Active participation in a scientific project can motivate citizens to become part of the solution to a given problem (Storey et al. 2016, Weeser et al. 2018), such as preventing the more frequent shifts of perennial rivers to intermittent rivers and ephemeral streams under climate change (Döll and Schmied 2012). On the contrary, Peters and colleagues (2015) showed that a lack of consideration for incorporating crowdsourced data into decision-making process in New Zealand resulted in demotivated citizens who stopped contributing to environmental monitoring.

The constant communication needed to promote the use of smartphone apps to the general public remains a challenging exercise, especially when the citizen science project covers several countries and aims to last for decades. Declining activity from the general public through time is a trend observed in several citizen science projects (Sauermann and Franzoni 2015), meaning that scientific project leaders have to multiply their efforts to keep people engaged. General communication—usually in English—through the project website, press releases, a YouTube channel, and through social media such as Twitter and Facebook is a first step. However, these media mainly reach the scientific community or people who are already concerned about environmental challenges. Therefore, national communication campaigns in local languages, targeting specific groups of users, such as fisher associations, nature protection organizations, kayaker groups, and pupils and students (through science festivals), has proven very useful to promote the use of smartphone apps. Besides direct communication with stakeholders, interviews given to national TV channels or to national radio stations are the most

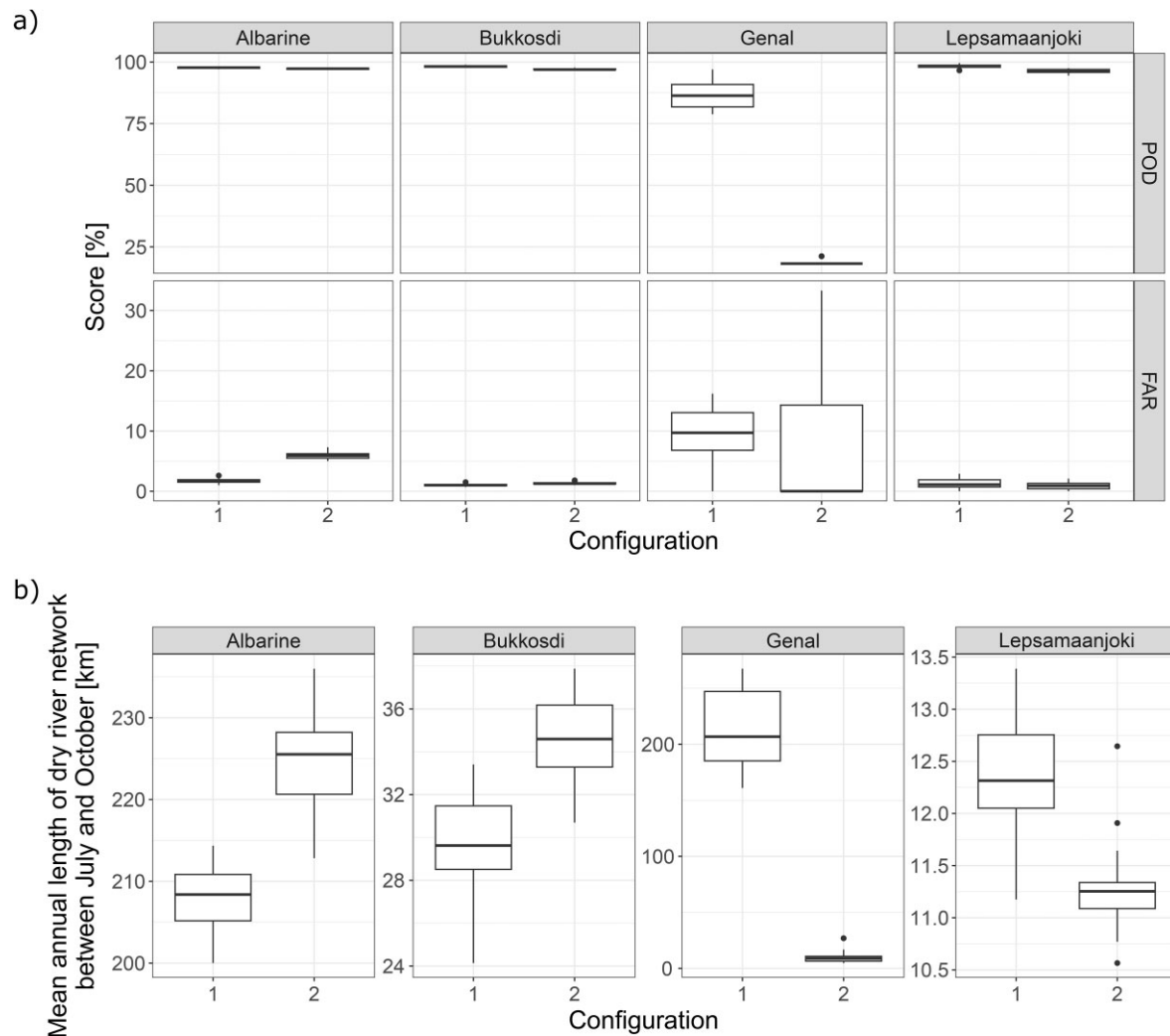


Figure 5. (a) The performance of the model using two different training data sets: with (configuration 1) and without (configuration 2) crowdsourced observations. The probability of prediction of dry events (POD) is calculated as the number of observed dry events that were correctly simulated over the total number of observed dry events. The false alarm ratio (FAR) is calculated as the number of observed flowing events that were wrongly simulated (i.e., simulated as dry) over the total number of simulated dry events. (b) The mean annual length of dry river network between July and October simulated with (configuration 1) and without (configuration 2) crowdsourced observations. The boxplots represent all 20 runs made with the model (see supplement S1 for further details on the modeling method).

successful events; the number of downloads greatly peaked shortly afterward (as seen in other projects; Sauermann and Franzoni 2015), showing people's interest in the app (e.g., for DRYRivers, 110 downloads on a single evening right after a 2-minute long interview for a national Hungarian TV channel, 66 downloads on a single day right after a 2-minute long interview for a Czech regional TV channel, 19 downloads in one afternoon right after a national radio program in France). These events are usually held when the smartphone app is released or within the time frame of a given research project, whereas communication around such apps should be maintained on the long term to ensure the use of the app over the billions of other apps available on the market. In addition, monitoring the number of active users is important, but it is even more important to keep these active users recording. Three main kinds of strategies have been developed so far: collecting points and badges as in DRYRivers and Crowdwat; communicating the scientific outcomes of the app; and gaming—for example, the comparison of two pictures that trains the user identifying erroneous observations (the Crowdwat game; Strobl et al. 2019), an approach also used for quality control.

Crowdsourced data is commonly perceived as less accurate than traditional scientific sources (e.g., academic, government) because of the engagement of people not trained as scientists. In the case of hydrological observations, typical mistakes consist of adding a new site close to an already existing site or marking a lake or a pond as a disconnected pool. These mistakes could be especially critical when using the crowdsourced data for hydrological modeling. Nonetheless, studies have reported that crowdsourced data can be as accurate as expert data (Engel and Voshell 2002, Cooper et al. 2014), especially when quality control measures are in place (e.g., gaming for Crowdwat or photo traps and flagging, currently under development for DRYRivers; Wiggins and Crowston 2011, Freitag et al. 2016).

By compiling all the available and accessible data, we hope to build a real-time map of flow intermittence in Europe within the next 5 years to complement the current fragmentation atlas (AMBER project, <https://amber.international/european-barrier-atlas>). The map would allow users to link flow intermittence with different types of barriers and download the data, and it could become the cornerstone of future, harmonized water

management strategies that are not limited to specific types of rivers or constrained by administrative boundaries.

Supplemental material

Supplemental data are available at [BIOSCI](https://doi.org/10.1093/biosci/biac098) online.

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